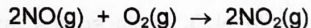


21 Nitrogen dioxide, NO_2 , is a pollutant. It is produced when nitrogen monoxide, NO , is oxidised.



The following data are obtained from an initial rates study into the kinetics of the reaction.

experiment	initial concentration of $\text{NO} / \text{mol dm}^{-3}$	initial concentration of $\text{O}_2 / \text{mol dm}^{-3}$	initial rate $/ \text{mol dm}^{-3} \text{ min}^{-1}$
1	0.80	0.80	0.020
2	0.80	0.32	0.0080
3	1.6	0.96	0.096
4	0.40	0.16	W

What is the value of W?

A 0.0004

B 0.001

C 0.002

D 0.004